WEEK 3 · LESSON 15 OF 24 · HANDOUT

AI for Spreadsheets and Data

Study time: **about 50 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=JYtZ2zsdE_s — open the lesson online to watch inside the course.

Your goal in this lesson: Use AI to write and explain spreadsheet formulas, clean and analyse simple data, and present findings — while checking every result.

Why this is a career superpower

Almost every job uses spreadsheets: budgets, stock lists, attendance, sales, grades. Many people lose hours because they do not know the right formula. AI can **write the formula, explain it, and help you fix errors** in seconds.

Writing formulas with AI

Describe your columns and what you want:

> *"In Google Sheets, column A has product names, column B has quantity sold, column C has price in RWF. Write a formula for column D that calculates total sales, and a formula that adds up all of column D."*

Then ask: *"Explain how this formula works, step by step."* Understanding the formula means you can fix it later.

Common tasks AI helps with

Task	Example prompt
Totals and averages	"Formula for the average score in B2:B40, ignoring empty cells."
Conditions	"Mark 'Pass' if the score in C2 is 50 or more, otherwise 'Fail'."
Look-ups	"Find a student's phone number from the 'Contacts' sheet using their ID." (VLOOKUP / XLOOKUP)
Counting	"Count how many customers are from Kigali in column E."
Cleaning	"Names are in UPPERCASE with extra spaces. Formula to make them tidy."
Errors	"My formula shows #N/A. Here it is: [formula]. Why, and how do I fix it?"

Analysing data

You can paste small, **non-confidential** tables (or upload files in tools that allow it) and ask:

- *"What are the 3 main trends in this sales data?"*
- *"Which month had the highest sales, and by how much compared with the average?"*
- *"Suggest the best chart to show this to a manager, and explain why."*

Many assistants now run real calculations on uploaded files. Still, **re-check key numbers** yourself — a misread column changes everything.

From numbers to a story

Data only matters when it leads to a decision. Ask AI to help you write the story:

> *"Here are the findings: [findings]. Write 3 short sentences for my manager: what happened, why it matters, and what we should do next."*

Checking rules for data work

- Test the formula on a few rows you can calculate by hand.
- Check totals against the original source.
- Never paste personal data such as names with phone numbers, salaries or ID numbers into a public AI tool — use made-up sample data to get the formula, then apply it yourself.

Remember this: AI turns spreadsheet problems into plain-language requests — but you test the formulas and check the numbers.

Name: _____ Date: _____

Activity

Task 1

Create a small sample table (10 rows) — for example a shop's sales or a class's marks — using **made-up** data.

Task 2

Ask AI for formulas: a total, an average, a Pass/Fail or Paid/Unpaid column, and a count of one category.

Task 3

Test each formula on two rows by hand.

Task 4

Ask AI for the best chart and a 3-sentence summary for a manager. Create the chart in your spreadsheet.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What is the best way to get a correct spreadsheet formula from AI?
   - A. Ask 'give me a formula'
   - B. Describe your columns and exactly what you want calculated
   - C. Ask for the longest formula
   - D. Upload a photo of your screen only
2. Why ask AI to explain a formula step by step?
   - A. To change the spreadsheet colour
   - B. Because formulas must be explained by law
   - C. To make it longer
   - D. So you understand it and can fix it later
3. Which function family is used to find a value from another table, such as a phone number by student ID?
   - A. AVERAGE
   - B. VLOOKUP / XLOOKUP
   - C. SUM
   - D. UPPER
4. Before trusting a new formula for a whole column, you should…
   - A. Ask a different AI
   - B. Delete the original data
   - C. Make the font bold
   - D. Test it on a few rows you can calculate by hand
5. You need a formula for a sheet with staff names, phone numbers and salaries. What is the SAFE approach?
   - A. Email the sheet to the AI company
   - B. Use made-up sample data to get the formula, then apply it to the real sheet yourself
   - C. Post it on social media for help
   - D. Paste the real sheet into a public AI tool
6. What turns data findings into something useful for a manager?
   - A. A longer table
   - B. Hiding the chart
   - C. A short story: what happened, why it matters, what to do next
   - D. More decimal places

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